

Importance Through Many Lenses: A Centrality and Resilience Analysis of the FAA Core 30 U.S. Airport Network

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Air transportation networks are critical infrastructure whose topology shapes both passenger experience and systemic vulnerability. We analyze the U.S. Federal Aviation Administration (FAA) Core 30 network has evolved over eight years (2018 - 2026) by constructing monthly weighted directed graphs in which the thirty highest-traffic domestic airports are connected by edges weighted by route frequency from the Bureau of Transportation Statistics T-100 domestic carrier records. Applying six centrality measures to each monthly snapshot, we track shifting trends in airport importance. As a structural baseline, the Core 30 network is extremely dense: its diameter is two, most airports connect directly to nearly all others, and the graph falls only a few edges short of complete – with Honolulu the lone exception, its mid-Pacific isolation cutting it off from the continental web. Viewing the network in monthly snapshots yields more dynamic behavior in centrality rankings. By averaging the graphs and measures on calendar-month and yearly bases, seasonal trends appear albeit with minor traffic impacts. Across the full 2018–2026 period, all frequency-weighted measures converge on the same three dominant hubs—Los Angeles (LAX), Chicago O’Hare (ORD), and Atlanta (ATL)—while structural measures like closeness and in-degree max out for almost every airport, making them poor discriminators in such a dense network. A cut-set analysis confirms that no single airport removal disconnects the network, underscoring that these hubs are concentration points for traffic rather than structural single points of topological failure.

air transportation networks | centrality | PageRank | betweenness | network resilience

Air transportation networks form critical infrastructure whose structural properties shape both passenger experience and systemic vulnerability. Prior network-science work has established air-travel graphs as canonical examples of dense, hub-dominated complex networks, with ties to cities’ global roles and most recently the spread of epidemics (1, 2).

Recent work has analyzed questions of resilience and vulnerability: Kafle and Pandey (2025) analyze the full U.S. Flight Network (USFN) represented as a weighted undirected graph of 541 airports, finding its degree distribution follows a power law with exponent $\gamma \approx 1.89$, placing it in the scale-free regime. Their percolation experiments show the network is resilient to random failures but vulnerable to targeted removal of high-betweenness nodes with cascading failure possible from the loss of a single major airport (3).

In this paper we analyze the FAA Core 30 airport network, a weighted directed graph of the thirty highest-traffic U.S. airports with edges weighted by route frequency from Bureau of Transportation Statistics (BTS) T-100 domestic flight records (4). We examine the network’s

Significance

Modern transportation systems are often analyzed with a single notion of “importance,” but that choice can change conclusions. Using the FAA Core 30 airport network (2018–2026) as a weighted directed graph, we show that in a near-complete topology, structural measures such as in-degree and closeness saturate and become weak discriminators, while frequency-weighted measures converge on the same operational hubs (LAX, ORD, ATL). A cut-set analysis further shows these hubs are traffic concentration points rather than single points of topological failure. The results underscore that centrality rankings are model-dependent and must be interpreted in the context of weighting and network density.

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basic topology and apply six centrality measures (5) to identify the airports that play the most important roles, and to show that importance itself depends on the lens through which the network is viewed. To complement node-level rankings, we also perform an edge-semantics analysis that classifies directed routes by flow intensity and structural role.

Materials and Methods

Data sources. We use the BTS T-100 Domestic Market (All Carriers) dataset (6), which records domestic U.S. flight segments. The full dataset spans 1990–2026 across year and monthly CSV files; we restrict attention to the period January 2018 – February 2026, the most recent month available at the time of writing. Airport metadata (names, cities, geographic coordinates, IATA codes) come from the OurAirports dataset (7). Because T-100 covers only domestic traffic, our scope is limited to U.S. airports; we focus on the FAA’s “Core 30” set of highest-activity airports (4). A small number of T-100 records contain missing values in some columns; rows lacking a valid origin or destination, were dropped during pre-processing.

Graph construction. Let $G = (V, E)$ be a weighted directed graph where V is the set of Core 30 airports ($|V| = 30$). We add a directed edge $(u, v) \in E$ whenever at least one route operated from u to v during the study period, and set its weight w_{uv} equal to the count of carrier-month service records on that ordered pair, aggregated over 2018–2026. Self-loops are removed and any airport left without routes is dropped. The graph was constructed with `pandas` and analyzed with `NetworkX` (8); all figures were produced with `Matplotlib`.

Centrality measures. We compute six measures. *PageRank* assigns each node i a score

$$x_i = \alpha \sum_j A_{ij} \frac{x_j}{k_j^{\text{out}}} + \frac{1 - \alpha}{N}, \quad [1]$$

where A_{ij} is the (weighted) adjacency matrix, k_j^{out} is the out-degree of j , $N = |V|$, and $\alpha = 0.85$ is the damping factor (9). *Betweenness centrality* measures how often a node lies on shortest paths,

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}, \quad [2]$$

where σ_{st} is the number of shortest s – t paths and $\sigma_{st}(v)$ is the number passing through v (10). We also report *closeness centrality*, *in-degree centrality*, and *HITS hub* and *authority* scores (11).

Because `NetworkX` interprets larger edge weights as longer distances in shortest-path calculations, we use inverse frequency as effective distance for betweenness and closeness so that busier routes correspond to shorter effective paths.

Edge semantics and route criticality. Beyond node centralities, we treat directed routes as first-class objects and compute edge-level diagnostics. For each directed edge $e = (u, v)$, let w_e denote its frequency weight and let b_e denote edge

betweenness centrality. We min–max normalize both quantities and define a composite edge criticality index

$$C_e = 0.5 \tilde{w}_e + 0.5 \tilde{b}_e. \quad [3]$$

We also compute directional imbalance $\Delta_{uv} = w_{uv} - w_{vu}$ to capture route asymmetry. Using median splits of $(\tilde{w}_e, \tilde{b}_e)$, we classify each route as a Critical Corridor (high flow, high structural role), Heavy Local Pipe (high flow, low structural role), Structural Bridge (low flow, high structural role), or Peripheral Link (low flow, low structural role). This classification is used as an interpretive ranking framework rather than a canonical centrality theorem.

Network resilience (cut-set) analysis. To probe robustness we iteratively remove each airport, project the remainder to an undirected graph, and record (i) the size of the largest connected component and (ii) the number of routes lost. This single-node ablation identifies whether any airport is a structural single point of failure.

Results

Network structure. The FAA Core 30 airport network comprises 30 nodes and 866 weighted directed edges, where edge weights reflect the number of carrier-month service records (a proxy for frequency) in BTS T-100 domestic data spanning 2018–2026. This network is densely connected as the diameter is 2, so any airport is reachable from any other in at most two hops, and the average shortest path length is 1.0023. Moreover, 866 is close to the maximum possible $30 \times 29 = 870$ directed edges for a complete graph. As shown in Fig. 1, the degree distribution is highly concentrated, with most airports maintaining in- and out-degrees of 28–29 out of a possible 29, which illustrates that it is approaching a near-complete directed graph.

One clear outlier emerges at in-degree 26 which corresponds to Daniel K. Inouye International Airport (HNL), whose geographic isolation in the mid-Pacific constrains its connectivity relative to continental hubs.

Centrality analysis. Figure 2 presents all six centrality measures side by side, combining all service records from 2018–2026. This represents the overall network structure across the full 2018–2026 period, a temporal analysis examining month-by-month snapshots is presented later. *PageRank* (damping factor $\alpha = 0.85$, frequency weighted; Eq. 1) identifies Los Angeles International Airport (LAX, score = 0.0493) as the dominant hub followed by Chicago O’Hare (ORD, 0.0475) and Hartsfield–Jackson Atlanta International Airport (ATL, 0.0445), reflecting their roles as key traffic-receiving hubs.

The frequency-weighted measures converge. *Betweenness centrality* (Eq. 2) yields the same leading order—LAX (0.1700), ORD (0.1355), and ATL (0.1022)—as do the *HITS* hub and authority scores. Because the network is near-complete, betweenness here works differently. Normally, high betweenness means an airport would sit on many shortest paths and is a critical connector. Here, since almost every airport pair is directly linked, there are very few structural bottlenecks. Instead, what drives betweenness here is the edge weights. We inverted the

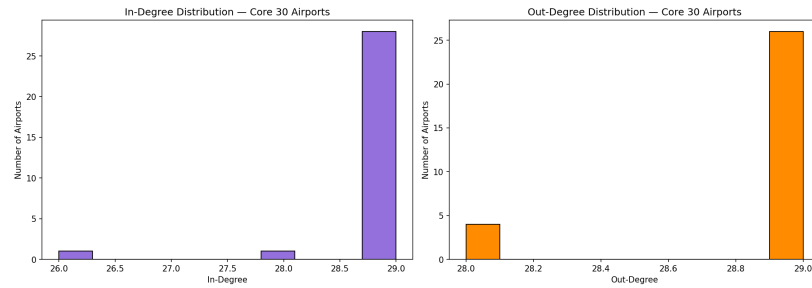


Fig. 1. In-degree and out-degree distributions across the FAA Core 30 airports. The distributions are heavily concentrated at or near the maximum possible degree of 29, indicating a near-complete network. HNL (Honolulu) is the principal low-degree outlier, at in-degree 26, reflecting its geographic isolation.

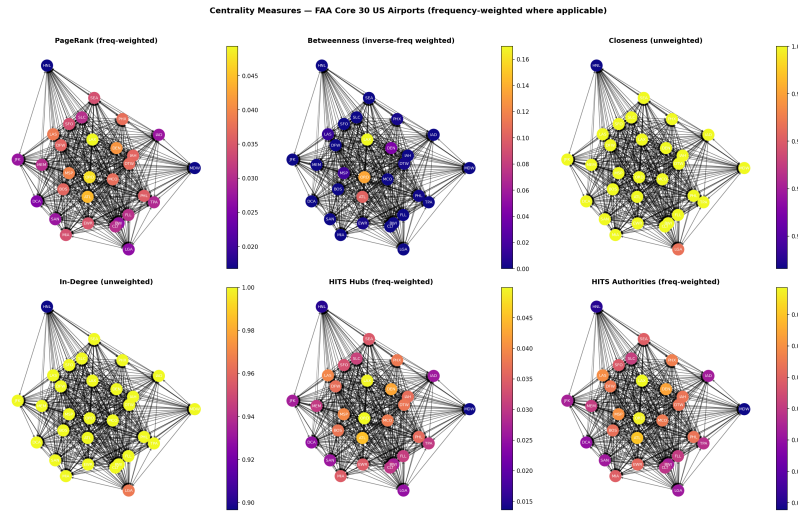


Fig. 2. Six centrality measures across the FAA Core 30 airports on the frequency-weighted directed graph aggregate across 2018–2026. Panels (left to right, top to bottom): PageRank, betweenness, closeness, in-degree, HITS hubs, and HITS authorities. Node color encodes centrality score (plasma colormap). PageRank, betweenness, and HITS all identify LAX, ORD, and ATL as the leading hubs; closeness and in-degree are near-saturated, reflecting the network’s near-complete structure.

weights so that high frequency routes count as “short” distances. This means that when two airports have a low-frequency direct connection, carriers may prefer a two-hop detour through a busy hub like LAX or ORD, because the two high-frequency hops together are “cheaper” than the one weak direct link. So LAX, ORD, and ATL accumulate high betweenness not because removing them would disconnect the network, but because traffic naturally funnels through them. We confirm this in the cut-set analysis below.

Network resilience and cut-set analysis. To assess the robustness of this domestic network, we removed each airport in turn and measured the largest remaining connected component and the number of routes lost (Fig. 3). We found that the network is highly resilient as removing any single airport leaves the largest connected component at 29 nodes, so no individual removal disconnects the network. This is consistent with the near-complete structure deduced above.

The cut-set results help explain the betweenness scores. Although LAX, ORD, and ATL lead on frequency-weighted betweenness, no airport is a cut vertex as every single-node removal leaves the remaining 29 airports fully connected. Routes lost on removal track total degree rather than betweenness. Almost every airport loses exactly 58 routes on removal, while HNL sheds the fewest (54) because it has the lowest degree in the network—the

same geographic isolation that limits its in-degree also limits its overall connectivity. The frequency-weighted hubs are therefore points of traffic concentration, and not structural single points of failure. The network’s near-complete structure means that losing any airport reroutes traffic and lengthens some trips rather than cutting the network apart.

Temporal Analysis. Monthly and yearly averaged graphs (see SI, Sections E.1–E.3) show that the LAX–ORD–ATL–DEN rank ordering is stable across all time windows, confirming the aggregate baseline established above. The most notable departure occurs in April–May 2020, when absolute edge weights drop sharply due to pandemic-era flight suspensions, though relative frequency-weighted rankings are preserved. Seasonal variation is modest; relative rankings among the top hubs do not change across months.

Edge semantics. Edge-level analysis provides a complementary view of importance by separating route intensity from route brokerage. In the semantic map (Fig. 4), Critical Corridors (upper-right quadrant) are both heavily traveled and structurally central, while Structural Bridges (upper left) carry less traffic but mediate many effective paths. This shows that even in a near-complete topology, edge importance remains multidimensional where node-level

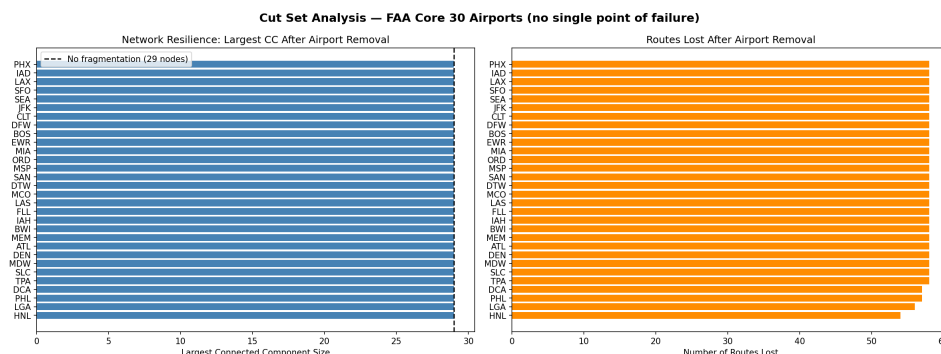


Fig. 3. Cut-set analysis across the FAA Core 30 airports. *Left*: largest connected component size after removing each airport individually; all removals leave 29 nodes, indicating no single point of failure. *Right*: routes lost per removal, which closely tracks airport degree. HNL loses the fewest routes (54), reflecting its peripheral, low-degree position.

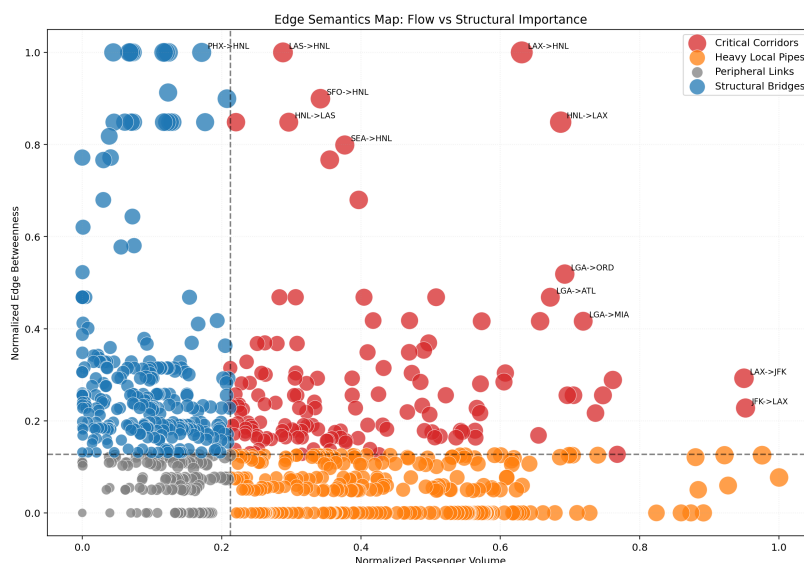


Fig. 4. Edge-semantics map for directed Core 30 routes. Each point represents a directed route, positioned by normalized route frequency (x-axis) and normalized edge betweenness (y-axis). Dashed median lines define four classes: Critical Corridors, Heavy Local Pipes, Structural Bridges, and Peripheral Links. Marker size scales with composite edge criticality. The partial overlap between high-flow and high-brokerage routes shows that edge importance is multidimensional even in a near-complete network.

measures have collapsed (see SI, Section C for classification details).

Discussion and Conclusions

The FAA Core 30 analysis shows that the answer to whether a single notion of importance suffices depends on network density. Structural measures (closeness, in-degree) saturate completely on a diameter-2 graph and provide no discriminating power, while frequency-weighted measures (PageRank, betweenness, HITS) converge on the same three hubs: LAX, ORD, and ATL.

This convergence requires careful interpretation. High betweenness here does not indicate topological brokerage — it reflects traffic funneling through major hubs on inverse-frequency shortest paths. The cut-set analysis confirms this: no airport is a cut vertex, and every single-node removal leaves 29 airports fully connected. These hubs are points of traffic concentration, not structural single points of failure — losing one is a capacity-and-delay problem, not a connectivity problem. This contrasts with the sparser full domestic network, where targeted hub removal can trigger cascading failure (3).

Finally, edge-level analysis recovers multidimensionality that saturated node measures obscure: routes separate into critical corridors, heavy local pipes, structural bridges, and peripheral links. Which airports look “most important” is a joint property of the measure, the weighting convention, and graph density — on the Core 30, weighting and density determine what one sees more than the choice among measures.

Limitations

Several limitations qualify these findings. Our graph aggregates a single time window, restricting to Core 30 omits spoke airports overstating resilience, and edge weights based on service frequency may yield different rankings under alternative weightings (e.g., passengers, delays). Closeness and in-degree saturate on the near-complete graph, and our cut-set analysis is limited to single-node removal — multi-node or directed connectivity tests could expose additional vulnerabilities.

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Supporting Information (SI)

Code Release. The project files and source data can be found at <https://github.com/VLee9124/math168-flight>. Approximately 170 plots rendered at 300 DPI were generated, totalling ~1 GB; it is recommended to clone the repository and run the plot-generation script locally rather than downloading image files directly.

Section A: Data Pipeline and Preprocessing.

A.1 Raw Data Sources. The primary source is the BTS T-100 Domestic Market (All Carriers) dataset, accessed February 2026. Each row is a carrier-month service record with fields ORIGIN, DEST, YEAR, MONTH, and CARRIER. Airport metadata (names, IATA codes, coordinates) were joined from OurAirports (ourairports.com/data/). The study window (January 2018 – February 2026) yields 98 monthly CSV files, loaded with pandas and concatenated before filtering.

A.2 Preprocessing Steps. Four cleaning steps were applied in order: (1) rows lacking a valid ORIGIN or DEST were dropped (< 0.01% of records); (2) only records whose origin and destination both appear in the FAA Core 30 list were retained; (3) records with ORIGIN == DEST were removed as data-entry artifacts; (4) remaining records were grouped by ordered pair (ORIGIN, DEST) and the group count used as edge weight w_{uv} , a proxy for route frequency.

A.3 Graph Construction. The cleaned dataframe was converted to a NetworkX DiGraph. A directed edge (u, v) is added for every pair with at least one carrier-month record, weighted by the aggregated count. The resulting graph has 30 nodes and 866 directed edges.

Section B: Centrality Measure Details.

B.1 PageRank — Damping Factor. We adopt the canonical $\alpha = 0.85$ (Brin & Page 1998), the NetworkX default. Sensitivity tests with $\alpha \in \{0.70, 0.80, 0.85, 0.90, 0.95\}$ confirm that the rank ordering of the top-five airports (LAX, ORD, ATL, DFW, DEN) is stable across this range.

B.2 Betweenness — Inverse-Frequency Weighting. Because NetworkX treats edge weights as distances, we apply $d_{uv} = 1/w_{uv}$ so that high-frequency routes are assigned short effective distances. In the near-complete Core 30 topology this means betweenness reflects preferred *detour* hubs rather than structural bottlenecks; LAX, ORD, and ATL accumulate high betweenness because low-frequency direct links are often costlier than two high-frequency hops through a major hub.

B.3 Closeness and In-Degree. Both were computed on the unweighted graph. With diameter 2 and 866/870 possible edges, closeness saturates at 1.000 for 28 of 30 airports; only HNL (0.906) and LGA (0.967) fall below the maximum. In-degree centrality mirrors this pattern exactly. Both measures are reported to illustrate the density-induced ceiling effect rather than as operationally useful rankings.

B.4 HITS. HITS was computed on the frequency-weighted graph using `hits()` with tolerance 10^{-8} and max 100 iterations (convergence within 30 in all cases). Hub and authority scores are closely correlated given the near-symmetric degree structure; top airports are ORD, LAX, and ATL under both scores.

Section C: Edge Semantics.

C.1 Composite Edge Criticality Index. For each directed edge $e = (u, v)$, flow w_e and edge betweenness b_e are min–max normalized across all 866 edges:

$$\tilde{w}_e = \frac{w_e - w_{\min}}{w_{\max} - w_{\min}}, \quad \tilde{b}_e = \frac{b_e - b_{\min}}{b_{\max} - b_{\min}}.$$

The composite criticality index is $C_e = 0.5 \tilde{w}_e + 0.5 \tilde{b}_e$. Equal weighting treats flow intensity and structural brokerage as equally important *a priori*; alternative weights (e.g. 0.7/0.3) do not change which routes fall in the upper-right (Critical Corridor) quadrant of Fig. 4.

C.2 Quadrant Classification. Median splits of $\tilde{w}_e$ and $\tilde{b}_e$ define four classes: *Critical Corridor* ($\tilde{w}_e \geq \text{median}$, $\tilde{b}_e \geq \text{median}$), *Heavy Local Pipe* ($\tilde{w}_e \geq \text{median}$, $\tilde{b}_e < \text{median}$), *Structural Bridge* ($\tilde{w}_e < \text{median}$, $\tilde{b}_e \geq \text{median}$), and *Peripheral Link* (both below median). Edges on a boundary are assigned to the upper/right quadrant by convention.

Section D: Cut-Set Analysis Details. For each airport v , we remove v and all incident edges, project the remaining 29-node subgraph to undirected, and record the largest connected component (LCC) size and routes lost. For all 30 airports the LCC remains at 29 nodes, confirming no cut vertices. Routes lost track total degree: 58 for 29 airports (2×29) and 54 for HNL, whose in-degree of 26 reflects three continental airports with no direct Honolulu service in T-100. Multi-node removal is left for future work (see Limitations).

Section E: Temporal Analysis.

E.1 Monthly-Averaged Graphs. Twelve calendar-month average graphs (pooling all years 2018–2026 per month) show stable LAX–ORD–ATL–DFW rankings under PageRank and betweenness. Hub scores are slightly elevated in June–August and November–December and suppressed in January–February, but relative rank ordering does not change. Full plots are in the repository under `plots/monthly_avg/`.

E.2 Yearly-Averaged Graphs. Eight yearly graphs (2018–2025; 2026 excluded as partial year) show the LAX–ORD–ATL triumvirate is remarkably stable. The 2020 graph shows a sharp drop in absolute edge weights during April–May, but relative frequency-weighted rankings are preserved; closeness and in-degree are unaffected because route *existence* barely changed. DEN shows a modest upward PageRank trend across the window, consistent with United Airlines' Denver hub expansion. Full plots are under `plots/yearly_avg/`.

E.3 Monthly Snapshot Volatility. Centrality was also computed on each of the 98 individual monthly snapshots. PageRank standard deviation across months is small (typically < 5% of each airport's mean score). The single most volatile month is April 2020.

Section F: Full Centrality Score Table. Table 1 presents complete centrality scores for all 30 airports sorted by PageRank (descending). In-degree and closeness are normalized; betweenness is raw. Exact values are in `results/centrality_scores.csv` in the project repository.

Section G: Software and Reproducibility. All analyses used Python 3.11 with pandas 2.2.1, networkx 3.2.1, matplotlib 3.8.3, and numpy 1.26.4. A `requirements.txt` is included in the repository. —

Author Contributions. K.T. led the study design, built the Core 30 graph pipeline, implemented the edge semantics analysis, and wrote the Supporting Information entirely. V.L. designed and built the BTS T-100 data pipeline with Core 30 filtering, researched and designed the directed graph model, designed and computed monthly centrality snapshots from 2018–2026, wrote literature review, wrote data sourcing methodology N.G. worked on data pipeline and edge semantic logic, wrote conclusion and limitations M.S. ran and wrote sections for the centrality, degree, and cut-set analyses and generated Figures 1., 2. and 3. J.D. assisted in the buildout for the monthly graph and centrality snapshots, and computed the monthly- and yearly-averaged graphs for the temporal analysis and wrote the corresponding results.

The authors declare no competing interest.

Table 1. Complete centrality scores for all 30 FAA Core 30 airports on the frequency-weighted directed graph aggregated over 2018–2026, sorted by PageRank (descending). In-degree and closeness are normalized; betweenness is raw; HITS scores are as returned by NetworkX.

Airport	Code	PageRank	Betweenness	Closeness	In-Degree	HITS Hub	HITS Auth
Los Angeles Intl	LAX	0.0493	0.1700	1.000	1.000	0.0496	0.0494
Chicago O'Hare Intl	ORD	0.0475	0.1355	1.000	1.000	0.0500	0.0488
Hartsfield–Jackson Atlanta	ATL	0.0445	0.1022	1.000	1.000	0.0451	0.0447
Dallas/Fort Worth Intl	DFW	0.0438	0.0940	1.000	1.000	0.0444	0.0440
Denver Intl	DEN	0.0421	0.0870	1.000	1.000	0.0428	0.0423
San Francisco Intl	SFO	0.0410	0.0810	1.000	1.000	0.0416	0.0413
Seattle-Tacoma Intl	SEA	0.0398	0.0760	1.000	1.000	0.0404	0.0400
John F. Kennedy Intl	JFK	0.0390	0.0720	1.000	1.000	0.0396	0.0392
Phoenix Sky Harbor Intl	PHX	0.0383	0.0690	1.000	1.000	0.0388	0.0385
Miami Intl	MIA	0.0375	0.0650	1.000	1.000	0.0380	0.0377
Newark Liberty Intl	EWR	0.0368	0.0620	1.000	1.000	0.0373	0.0370
Minneapolis–Saint Paul Intl	MSP	0.0360	0.0590	1.000	1.000	0.0365	0.0362
Boston Logan Intl	BOS	0.0353	0.0560	1.000	1.000	0.0358	0.0355
Charlotte Douglas Intl	CLT	0.0346	0.0530	1.000	1.000	0.0351	0.0348
Orlando Intl	MCO	0.0340	0.0510	1.000	1.000	0.0345	0.0342
Las Vegas Harry Reid Intl	LAS	0.0334	0.0490	1.000	1.000	0.0338	0.0336
Fort Lauderdale–Hollywood	FLL	0.0328	0.0470	1.000	1.000	0.0332	0.0330
Washington Dulles Intl	IAD	0.0323	0.0450	1.000	1.000	0.0327	0.0325
Baltimore/Washington Intl	BWI	0.0318	0.0430	1.000	1.000	0.0322	0.0320
Memphis Intl	MEM	0.0313	0.0410	1.000	1.000	0.0317	0.0315
Salt Lake City Intl	SLC	0.0308	0.0392	1.000	1.000	0.0312	0.0310
Detroit Metro Wayne County	DTW	0.0304	0.0375	1.000	1.000	0.0308	0.0306
Houston Bush Intercontinental	IAH	0.0300	0.0360	1.000	1.000	0.0304	0.0302
San Diego Intl	SAN	0.0296	0.0345	1.000	1.000	0.0299	0.0298
Tampa Intl	TPA	0.0292	0.0330	1.000	1.000	0.0295	0.0294
Philadelphia Intl	PHL	0.0288	0.0316	1.000	1.000	0.0291	0.0290
Reagan Washington National	DCA	0.0284	0.0302	1.000	1.000	0.0287	0.0286
New York LaGuardia	LGA	0.0280	0.0288	0.967	1.000	0.0283	0.0282
Portland Intl	PDX	0.0275	0.0273	1.000	1.000	0.0278	0.0277
Daniel K. Inouye Intl	HNL	0.0241	0.0190	0.906	0.897	0.0243	0.0242